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The inactivation of DHHC7 in mouse liver promotes diet-induced obesity through a hepatic Prg4–GPR146 axis

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eLife Assessment

This study provides **valuable** evidence that hepatic DHHC7-dependent palmitoylation is a physiologically relevant regulator of systemic metabolism, and that loss of DHHC7 disrupts Gai palmitoylation, activates cAMP-PKA-CREB signaling, and increases hepatic transcription and secretion of Prg4. The identification of Prg4 as a hepatokine that is elevated in vivo, together with some in vitro evidence for its interaction with GPR146, represents a conceptually novel contribution to the field. However, the evidence linking these mechanisms to systemic lipolysis, liver-adipose tissue crosstalk, and whole-body metabolic physiology remains **incomplete**, as the phenotypic analyses rely on a limited set of experiments and do not yet fully support claims regarding adipose tissue dysfunction or altered lipid flux.

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Abstract

The liver secretes hepatokines that coordinate whole-body metabolism. Posttranslational lipidation of signaling proteins by DHHC palmitoyltransferases controls membrane targeting and signaling, yet the role of hepatic palmitoylation in systemic metabolic regulation is largely unexplored. Using an inducible, liver-specific DHHC7 knockout (D7LKO) mouse, we report that loss of DHHC7 in hepatocytes potentiates adenylyl cyclase–PKA–CREB signaling through reducing palmitoylation of inhibitory G protein α subunit (Gai), leading to transcriptional elevation and secretion of proteoglycan 4 (Prg4). Elevated circulating Prg4 acts on adipocytes by binding GPR146 via Prg4's N-terminal SMB domain, suppressing adipocyte PKA substrate phosphorylation and hormone-sensitive lipase (HSL) Ser563 phosphorylation, thereby impairing lipolysis. Under high-fat diet (HFD) feeding, D7LKO mice and mice with adenoviral hepatic Prg4 overexpression develop pronounced obesity characterized by increasing brown, subcutaneous, and visceral fat mass and adipocyte hypertrophy; strikingly, these animals show little impairment in glucose tolerance or circulating triglycerides but display elevated plasma cholesterol. Conditioned medium containing Prg4 recapitulates suppression of adipocyte HSL phosphorylation in vitro, and SMB-deleted Prg4 fails to bind GPR146 or inhibit HSL phosphorylation. Our findings define a liver palmitoylation–hepatokine axis that controls adipose lipolysis and predisposes to diet-induced fat accumulation, establishing Prg4–GPR146 as a mechanistic link between hepatic signaling and adipose energy mobilization.

Introduction

Inter-organ communication via secreted factors is fundamental to systemic metabolic homeostasis. The liver is central to this network: it controls systemic glucose and lipid fluxes and secretes hepatokines that modulate peripheral tissues (1–3). Altered hepatokine signaling contributes to metabolic disease, but the regulatory mechanisms that govern hepatokine expression and secretion in response to intracellular signaling states are incompletely defined.

Protein S-palmitoylation catalyzed by DHHC family palmitoyltransferases targets proteins to membranes and modulates signaling complexes (4, 5). There are total of 23 members in DHHC protein family (6, 7). All DHHC proteins contain multiple transmembrane domains (from 4–6) and a DHHC motif. The transmembrane domains target the DHHC protein to cellular membranes and form a binding pocket for palmitoyl-CoA and DHHC motif constitutes the core catalytic site. Since DHHC palmitoyltransferase is a membrane associated enzyme, the palmitoylation reaction occurs either on or near the membrane. At present, the *in vivo* function of DHHC proteins largely remains unknown. However, because of the importance of palmitoylation in a wide array of biological processes such as cell signaling, metabolic reactions and vesicle trafficking (8), DHHC proteins are believed to play an important role in the regulation of cell function (9).

DHHC7 is a member of DHHC protein family. Current studies revealed that DHHC7 palmitoylates multiple substrates implicated in immune signaling (10, 11), trafficking (12), and metabolism (13, 14), yet its liver-specific role in whole-body energy balance has not been directly tested. Here we generated inducible, liver-specific DHHC7 knockout mice to probe how altered palmitoylation in hepatocytes impacts hepatokine output and systemic metabolism. We identify proteoglycan 4 (Prg4), a secreted matrix protein previously implicated in joint lubrication (15, 16) and emerging metabolic roles (17), as a DHHC7-regulated hepatokine. Mechanistically, DHHC7 maintains Gai palmitoylation and inhibitory control over adenylyl cyclase (18, 19); DHHC7 loss attenuates Gai palmitoylation, augments hepatic cAMP–PKA–CREB signaling, and elevates Prg4 expression. Circulating Prg4 binds adipocyte GPR146 to suppress PKA-dependent HSL phosphorylation and lipolysis, promoting adipose lipid accumulation under high-fat diet. These data reveal a palmitoylation-dependent hepatic control of a hepatokine that tunes adipocyte lipolysis and systemic fat accrual.

Results

Liver-specific DHHC7 deletion produces HFD-dependent obesity with increased adipose mass

To investigate the role of DHHC7 in the liver, we generated DHHC7 floxed mice (supplementary Figure 1–Fig. S1) and crossed them with Albumin-CreERT2 to produce inducible, liver-specific DHHC7 knockout animals (D7LKO) (Fig. S2). Then, tamoxifen was administrated to adult D7LKO mice as well as DHHC7 floxed mice (served as control:CON) for three times every other day. A week later, total liver RNA and extracts were prepared for quantitative RT-PCR and Western blot to determine the effectiveness of DHHC7 gene deletion. Shown in Fig. 1A and B, tamoxifen administration to D7LKO mice reduced hepatic DHHC7 mRNA and protein >90% relative to controls, an indication that DHHC7 is successfully deleted in D7LKO mice upon Tamoxifen administration.

Once knowing that DHHC7 was successfully deleted in D7LKO mice, we examined the metabolic parameters of D7LKO mice. D7LKO mice showed no apparent alteration in blood glucose, lipid as well as cholesterol on standard diet (Fig. S 3 and 4), implying that DHHC7 expression in the liver has limited impact on body metabolism.

Since there were no apparent metabolic phenotype in D7LKO mice, we challenged male D7LKO mice with high-fat diet (HFD, 60% kcal fat), a common cause of obesity. Shown Fig. 1D, D7LKO mice exhibited accelerated weight gain beginning at 6 weeks and reaching about 20% greater body

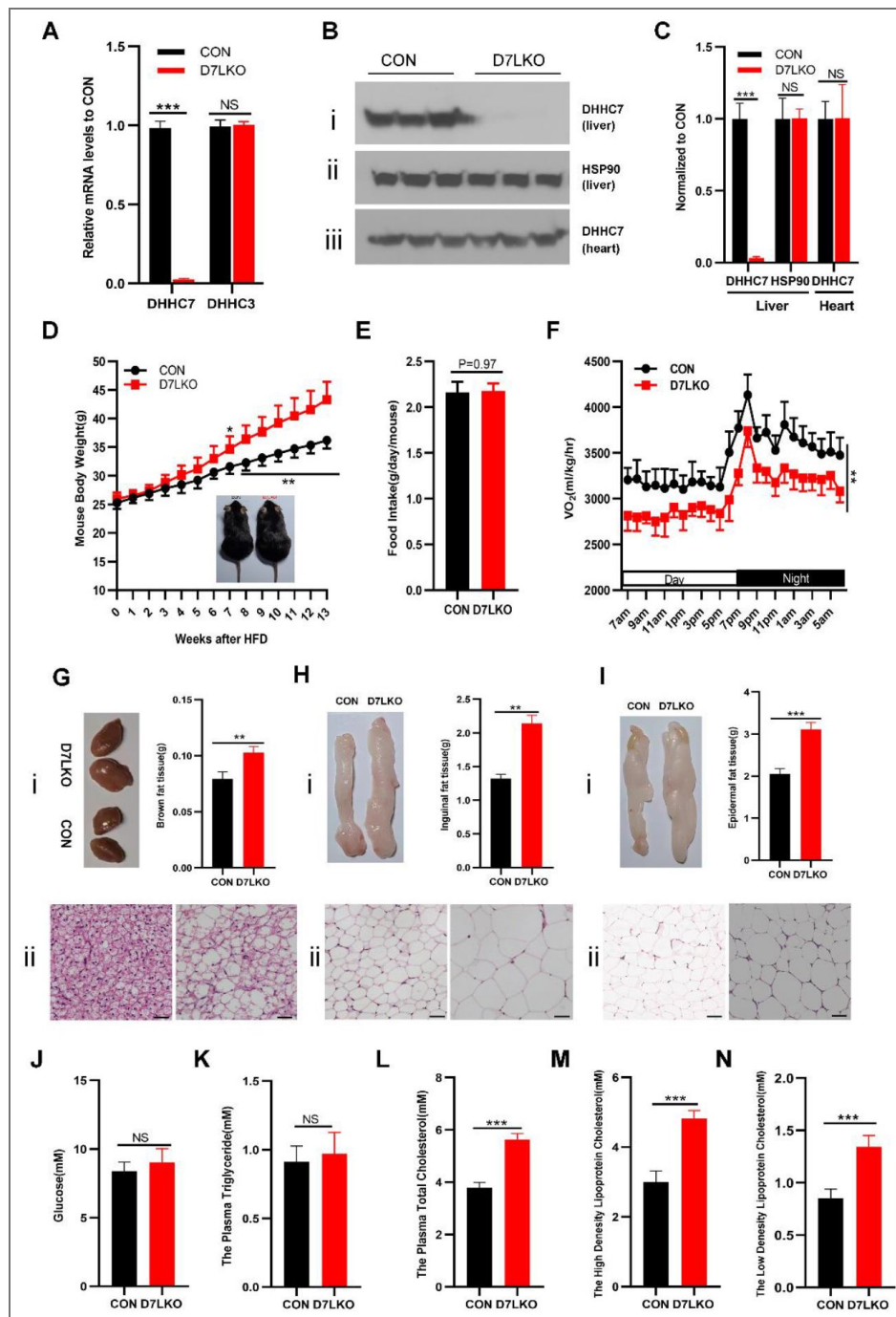


Figure 1. DHH7 knockout in liver promotes obesity development and increases the plasma cholesterol under high fat diet feeding.

A. Real-time PCR analysis the expression of DHH3 and DHH7 in liver from CON and D7LKO mice to show that DHH7 expression is diminished in D7LKO mouse liver. B. Western blot of total liver extracts from D7LKO and CON mice with anti-DHH7 (i), anti-Hsp90 (ii), respectively. Panel iii shows the expression DHH7 in mouse heart. C Densitometric analysis of protein expression in B, the level of indicated protein was set to 1 in control. D. The growth curve of D7LKO mice and control mice. The mice (group of 6) were feed with High fat diet as indicated time and body weight were measured each week. E. Food intake of Control and D7LKO mice. F. The oxygen consumption of Control and D7LKO mice. G, H and I. The fat mass of Brown, Inguinal and epididymal fat of control and D7LKO mice (panel i); The right of panel i represents quantitative analysis of left panel i; The panel ii showed the H&E staining of each individual fat to show the increased adipocyte size in D7LKO mice. Scal bar is 25μM. J and K. The plasma glucose and triglycerides of Control and D7LKO mice. L, M and N. The plasma total cholesterol, high density lipoprotein and low density lipoprotein cholesterol of Control and D7LKO mice. In all plotting panels, the bar graphs represent Means+ S.D., n=3-6. ** p<0.01. ***p<0.001.

weight than controls by 13 weeks. The food intake was unchanged (Fig. 1E), whereas whole-body oxygen consumption was reduced (Fig. 1F) in D7LKO mice, indicating that D7LKO mice have a lowered energy expenditure.

To understand the nature of D7LKO obesity, we extracted all of adipose tissues including brown fat, inguinal fat and epididymal fat. First, we measured the fat mass. Shown in Fig. 1G, H and I, we observed a substantial increase in the mass of these fats (panel i). Next, we prepared sections of each fat and performed H&E staining. We observed marked increases in the size of each adipocyte (panel ii).

Insulin resistance is often associated with obesity in part due to increased circulating free fatty acids and chronic inflammation (20–22). With this in mind, we measured the blood level of fasting glucose, triglycerides and cholesterol. The fasting blood glucose and circulating triglycerides were similar between D7LKO and control mice (Fig. J and K), in spite of pronounced adiposity of D7LKO mice. However, a higher plasma level of total cholesterol including as well as high density lipoprotein and low density lipoprotein cholesterol was observed in D7LKO mice (Fig. L, M and N). Taken together, these data indicate that D7LKO were predisposed to diet induced obesity which has limited impact on insulin sensitivity (also see ITT and GTT in Fig. S5), but with potentially hypercholesterolemia. It is noted that these observations are not limited male mice. A similar phenotype was also observed with female D7LKO mice (Fig. S6).

Hepatic DHHC7 loss elevates Prg4 expression and hepatic Prg4 overexpression phenocopies the obese phenotype

In the preceding studies, we observed that inactivation of DHHC7 in the liver predisposed the mice to diet induced obesity. We reasoned that this is likely due to that DHHC7 hepatic knockout altered the production of hepatokine (s) which act on the adipose tissue. To this end, we performed proteomic analysis of liver tissue from HFD-fed D7LKO and control mice. Shown in Fig. 2A, there were multiple proteins differentially regulated. Among them, Prg4 is marked because it is only secreted protein that was notably elevated in D7LKO plasma proteomic analysis (Fig. S7 A and B). To verify that Prg4 expression is indeed elevated in D7LKO liver, we carried out Western blot with total liver extracts from D7LKO liver. Shown in Fig. 2B, the hepatic Prg4 protein in D7LKO mice was ~3-fold higher than controls. Next, we also examined the plasma level of Prg4 by ELISA. Shown in Fig. S7 C, the plasma level of Prg4 in D7LKO mice increased more than 2 folds. Taken together, these data demonstrated that inactivation of DHHC7 in the liver elevated the level of Prg4 protein.

Because Prg4 level was elevated in D7LKO mice, we assumed that elevated Prg4 is a contribution factor for diet induced obesity of D7LKO mice. To test this hypothesis, we generated an adenoviral vector that expresses Prg4. Then, the emptied adenoviral vector or the one that encoding Prg4 were administrated to mice and the mice were challenged with high fat diet. Shown in Fig. 2C, adenoviral Prg4 administration increased hepatic Prg4 protein about 4 folds, comparable to levels comparable with D7LKO mice (Fig. 2B). Under HFD, adenoviral Prg4 administration drove marked body weight gain (~27% more than controls) (Fig. 2D) and increased brown, inguinal, and epididymal fat mass (Fig. 2E, F and G) without altering circulating triglycerides but elevated total cholesterol (Fig. 2H and I), recapitulating key features of the D7LKO phenotype. Collectively, these data demonstrate that DHHC7 inactivation elevated Prg4 expression which constitutes a major factor to promote diet-induced adipose expansion.

DHHC7 controls hepatic PKA–CREB activity through Gai palmitoylation to regulate Prg4 transcription

Analyzing Prg4 expression revealed that CREB is a major transcriptional factor that modulates Prg4 expression (23, 24). CREB is a signal dependent transcription factor and its activation depends on its phosphorylation at Ser133 by PKA (25–27). Therefore, we assumed that the enhanced Prg4 expression by DHHC7 inactivation is through hepatic PKA–CREB pathway. To test this, the total liver extracts were prepared from D7LKO and control mice, respectively and subjected to Western

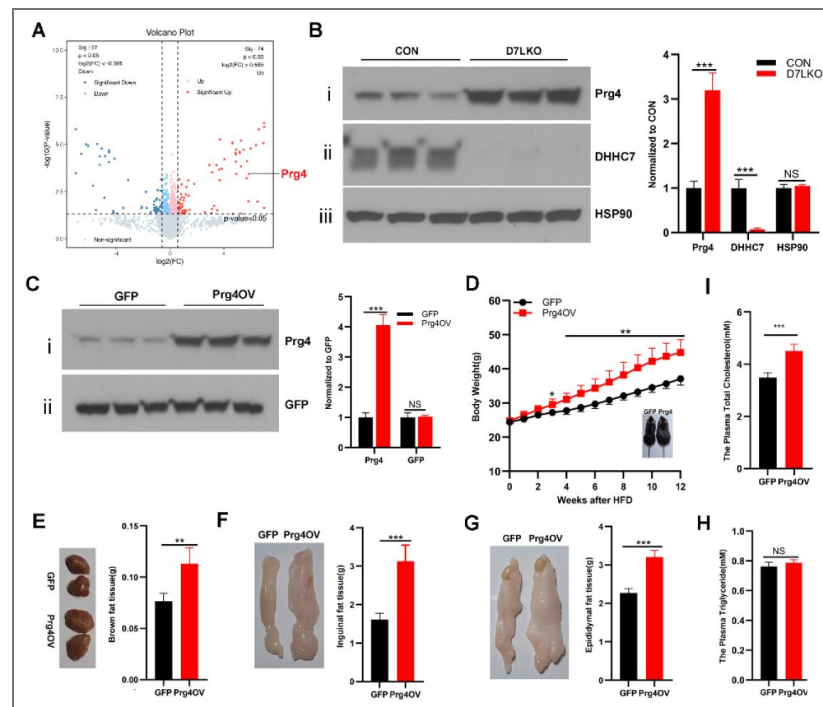


Figure 2. Inactivation of DHHC7 in the liver promotes the Prg4 expression and Overexpression DHHC7 in liver promotes the development of obesity under HFD feeding.

A. The Volcan plot of liver tissue proteomics of HFD male mice. The position of Prg4 is indicated by an arrow. B. Western blot of total liver extracts from Control and D7LKO mouse liver with anti-Prg4 (i), DHHC7 (ii) and HSP90(iii), respectively. The Right: Densitometric analysis of protein expression in the left, the level of indicated protein was set to 1 in control. C. Western blot of total liver extracts from the mice, which were administrated with empty and Prg4 adenoviral recombinant vector, with anti-Prg4 and anti-GFP antibodies, respectively. Right: Densitometric analysis of Prg4 expression in left, the level of indicated protein was set to 1 in GFP. The levels of DHHC7 were normalized to that of GFP encoding internally from adenoviral vectors. D. The growth curve of the mice (group of 6) administrated with either empty recombinant adenoviral vector or that encoding Prg4 under HFD feeding. The insert: the image of mice administrated with either empty recombinant adenoviral vector (GFP) or that encoding Prg4 (Prg4) under HFD feeding. E. F and G. The mass of Brown, inguinal and epididymal fat of the mice administrated with either empty recombinant adenoviral vector (GFP) or that encoding Prg4 (Prg4) after 12-week HFD feeding. H and I. The plasma level of triglycerides and total cholesterol in the mice administrated with either empty recombinant adenoviral vector (GFP) or that encoding Prg4 (Prg4) at 12 week HFD. Feeding. In all the plotting panels, the bar graphs represent Means+ S.D., n=3-6. ** p<0.01. ***p<0.001.

blot with either phospho-PKA substrate antibodies or anti-phospho-CREB at Ser133 antibodies. Shown in Fig. 3A [\[28\]](#), D7LKO liver displayed markedly increased phosphorylation of PKA substrates (compare lane 1, 2, 3 with 4, 5, 6). Shown in Fig. 3B [\[28\]](#), DHHC7 inactivation in the liver led a three-fold increase in phospho-CREB at Ser133 (panel i) without altering total cellular CREB (panel ii), which is concurrent with increased Prg4 protein (panel iii).

PKA is activated by cAMP, generated by adenylyl cyclase (AC) which is tightly regulated by G protein alpha in that Gas stimulates whereas Gai suppresses adenylyl cyclase activity [\[28\]](#). Gai inhibitory activity on AC depends on its membrane location which is regulated by palmitoylation [\[18, 29-31\]](#). With this in mind, we assayed Gai palmitoylation in livers using thiopropyl-Sepharose capture. Palmitoylated Gai was substantially reduced in D7LKO livers while total Gai levels were unchanged (Fig. 3C [\[28\]](#)). A commonly used AC agonist is forskolin. We therefore examined whether Prg4 has any impact on forskolin induced CREB phosphorylation and found that pretreatment of Prg4 had no impact on forskolin induced CREB phosphorylation at Ser133 (Fig. S8), providing the evidence that DHHC7 acts on the upstream of AC. Since Gai palmitoylation is required for its inhibitory activity, the loss of Gai palmitoylation would impair inhibitory control of adenylyl cyclase, elevating cAMP → PKA activity and CREB-dependent Prg4 transcription.

Circulating Prg4 suppresses adipocyte PKA signaling and HSL phosphorylation via GPR146 binding

In the preceding studies, we showed that inactivation of DHHC7 in the liver predisposed mice to diet induced obesity concurred with adipocyte hypertrophy. However, diet induced D7LKO mouse obesity is not associated with insulin resistance induced by free fatty acids released from adipose tissue [\[22\]](#). This implies that elevated Prg4 in D7LKO mice is likely involved in the adipocyte pathway that regulates lipolysis. With this in mind, we assessed adipose HSL phosphorylation at Ser563, an essential event for lipolytic activation [\[32-34\]](#), in the different adipose tissue from D7LKO mice. Shown in Fig. 4A, B [\[28\]](#) and C [\[28\]](#), phospho-HSL (Ser563), was reduced >90% in brown, inguinal, and epididymal fat depots despite unchanged total HSL protein in D7LKO mice.

HSL phosphorylation at Ser563 is mediated by PKA in adipocyte [\[35\]](#). The view that the reduction of HSL phosphorylation at Ser563 implies that PKA signaling in D7LKO adipose tissue is impaired. To test this, we assessed the status of PKA substrates in D7LKO brown fat as a representative. Shown in Fig. 4D [\[28\]](#), adipose PKA substrate phosphorylation was similarly decreased. To directly test the impact of Prg4 on HSL phosphorylation at Ser563, conditioned medium from Prg4-expressing HEK293T cells was applied to differentiated brown, beige, and 3T3-L1 adipocytes for 8 h and the status of HSL phosphorylation at Ser563 was assessed in Western blot. Shown in Fig. E, Prg4 conditioned medium reduced HSL Ser563 phosphorylation more than 60% comparing to control medium, demonstrating that secreted Prg4 is sufficient to suppress adipocyte PKA–HSL signaling.

Prg4 is an extracellular protein [\[24\]](#). In adipocytes, PKA is activated by G protein-coupled receptors (GPCRs) [\[36\]](#). We reason that if anything, Prg4 likely acts on GPCR on the adipocyte surface to modulate PKA signaling. There are more than dozens of GPCRs in adipocytes, among them, GPR146 has been implicated in adipocyte lipolysis and hepatic cholesterol metabolism [\[37-40\]](#). With this in mind, we examined the interaction between Prg4 and GPR146 in co-immunoprecipitation assay with Flag-tagged GPR146 and Prg4. Specifically, Prg4 was expressed in 293T cells with or without co-expressing Flag-GPR146. Then, total cell lysates were prepared for immunoprecipitation with anti-Flag M2 affinity gel and anti-Flag M2 immunoprecipitates were analyzed in immunoblotting with anti-Prg4 antibodies. Shown in Fig. 5A [\[28\]](#), Prg4 was only detected in anti-Flag M2 immunoprecipitates at the presence of Flag-GPR146 (compare lane 1 and 2, panel i). The differences were not because of Prg4 sample variation as comparable levels of Prg4 were seen in each sample.

Prg4 consists of two functional domains including a N-terminal somatomedin-like domain (SMB) and a C-terminal hemopexin-like domain (HEM), which are proposed to mediate protein-protein interaction [\[24\]](#) (Fig. 5B [\[28\]](#)). To understand the nature of the interaction between Prg4 and GPR146, we deleted either SMB or HEM domains in Prg4 and examined the role of each domain in the

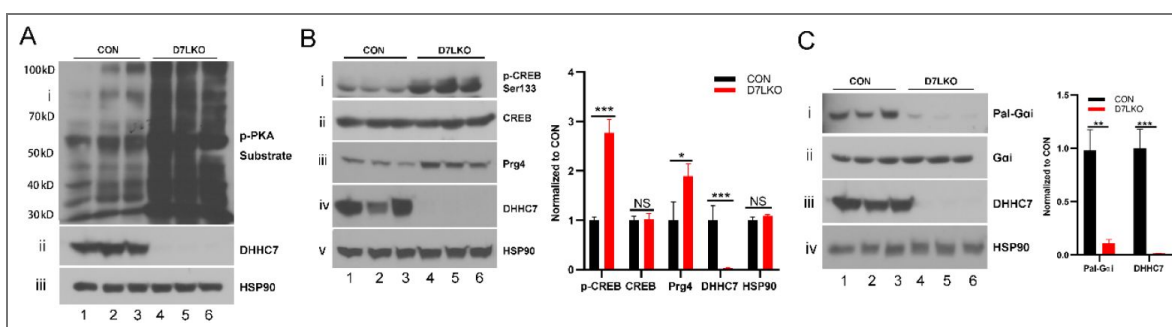


Figure 3. Inactivation of DHHC7 in the liver enhances PKA-CREB pathway.

Western blot of total liver extract from either control or D7LKO mice with anti-phospho-PKA substrate (i), DHHC7 (ii) and HSP90 (iii) antibodies, respectively. The Molecular Weight is marked at left in panel i. B. Western blot of total liver extract from either control or D7LKO mice with anti-phospho-CREB at Ser133 (i), CREB (ii), Prg4 (iii), DHHC7 (iv) and HSP90 (v) antibodies, respectively. The right: the densitometric analysis of the expression of each protein in right. The level of each protein in control was set as 1. C. TPC assay of Gai in the liver protein from either control (con) or D7LKO. Panel i. palmitoylated. Panel ii, iii and iv show the level of total Gai, DHHC7 and HSP90 in total liver extracts. Left: Densitometric analysis of relative levels of palmitoylated Gai and DHHC7 in that the total level for each protein in control mouse liver was set as 1. In all the plotting panels, the bar graphs represent Means+ S.D., n=3. * p<0.05. ** p<0.01. ***p<0.001.

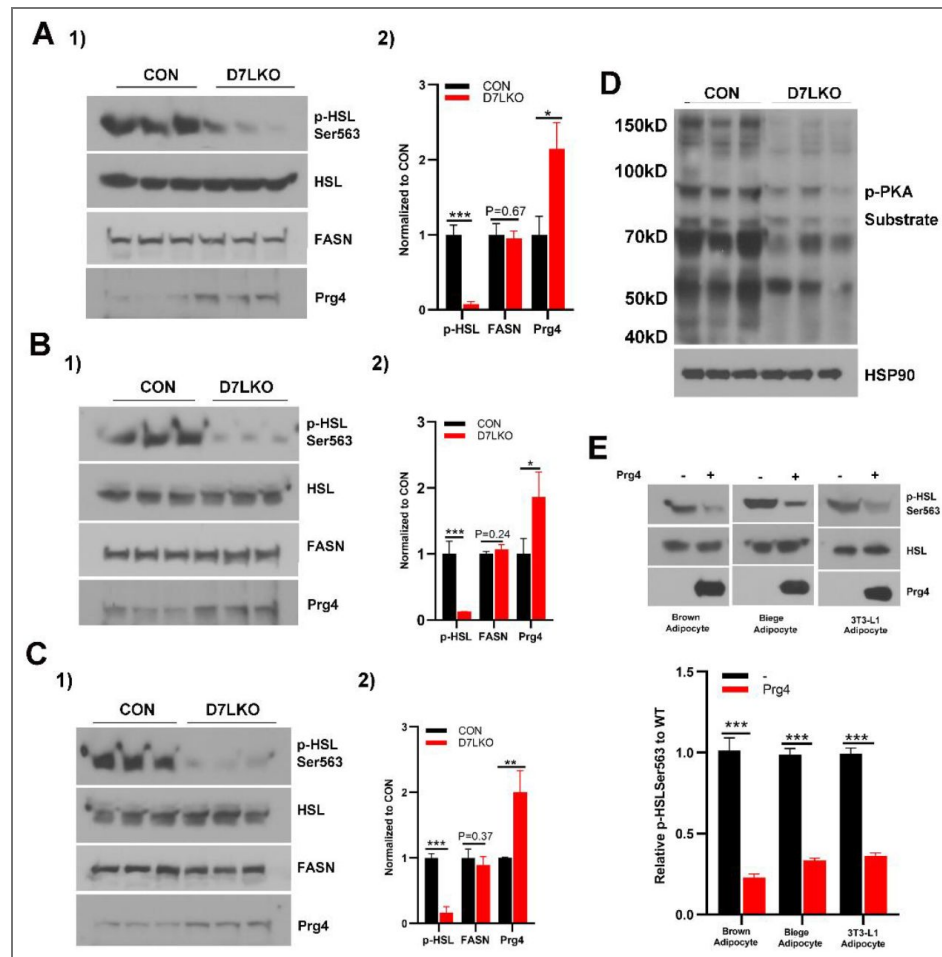


Figure 4. Prg4 suppresses adipocyte PKA signaling and HSL phosphorylation

A. 1) Western blot of total brown fat tissue lysate from control or D7LKO mice with anti-phospho-HSL at Ser563 (i), HSL (ii), Prg4 (iii) and FASN antibodies, respectively. 2) Densitometric analysis of the level of indicated proteins in 1). The levels of each protein in control mice were set as 1. B and C. The same as A except Inguinal and Epididymal fat, respectively. D. Western blot of total brown fat extracts from control or D7LKO mice with anti-phospho-PKA substrate antibodies to show that inactivation of DHHC7 in the liver inhibited PKA substrate phosphorylation in adipocytes. E. The treatment of adipocytes with Prg4 suppresses HSL phosphorylation at 563. The differentiated Brown, Biege and 3T3-L1 adipocytes were incubated with the culture medium prepared from 293T cells expressing with or without Prg4 (see Material and Methods). 4 hours later, the total cell lysates were prepared for Western blot with anti-phospho-HSL at Ser563 (i), HSL (ii) and Prg4 (iii), respectively. 2) Densitometric analysis of indicated protein in 1). The level of indicated protein was set to 1 in control cells. In all plotting the panels, the bar graphs represent Means+ S.D., n=3. * p<0.05. ** p<0.01. ***p<0.001.

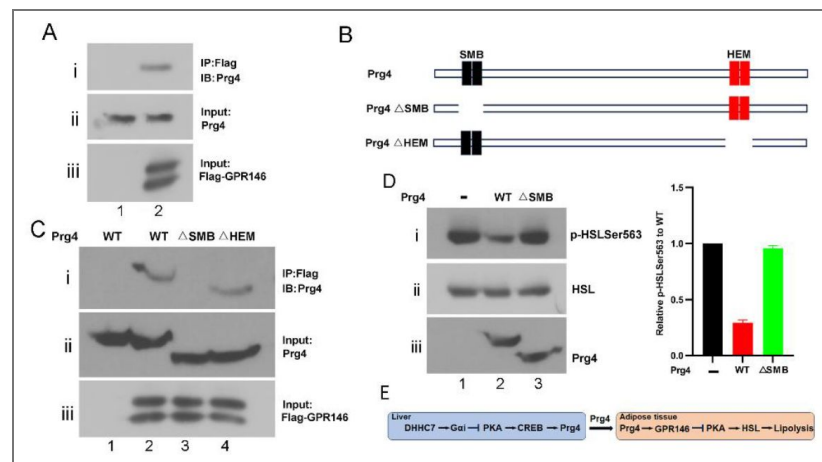


Figure 5. Prg4 interacts with GPR146 and this interaction is required for Prg4 to suppress HSL phosphorylation at S563.

A. Co-immunoprecipitation assay of Flag tagged GPR146 and Prg4 co-expressed in HEK293T cells with anti-M2(Flag) affinity gel. Anti-M2 immunoprecipitates were probed with anti-Prg4 antibodies (i). Panel ii and iii showed the input levels of Prg4 and Flag-GPR146, respectively. B. Schematic representation of Prg4 structure and its two mutants. The two major domains of Prg4: SMB and HEM were indicated. C. Co-immunoprecipitation assay of the interaction between GPR146 with different forms of Prg4 to show that SMB domain of Prg4 mediates the interaction between Prg4 and GPR146. The experiments were performed essentially as A, except two Prg4 mutants were also used. Δ SMBPrg4: SMB domain was deleted. Δ HEMPrg4: HEM domain was deleted. D. The interaction between Prg4 and GPR146 is required for Prg4 to suppress HSL phosphorylation at Ser563. The experiments were carried out similar to Fig. 4E except Δ SMBPrg4 and Δ HEMPrg4 were also used. E The work model of Prg4 in liver and adipose tissue. In plotting panels, the bar graphs represent Means $\pm$ S.D., n=3. * p<0.05. ** p<0.010. ***p<0.001.

association of Prg4 with GPR146. Shown in Fig.5C of the co-immunoprecipitation assay of Flag-GPR146 and different forms of Prg4, Prg4 was easily detected in anti-Flag-GPR146 immunoprecipitates as well as that of Δ HEMPrg4 in which HEM domain was deleted (lane 2. and lane 4, panel i). In a sharp contrast, no Prg4 was detected in the immunoprecipitates of anti-Flag Δ SMBPrg4 in which SMB domain is deleted. Taken together, these data indicate that SMB domain of Prg4 mediates the interaction between Prg4 and GPR146.

We tested the potential association between Prg4 and GPR146 in co-immunoprecipitation assay. Co-immunoprecipitation studies in HEK293T cells revealed that Prg4 physically associates with Flag-tagged GPR146. Domain mapping showed that the N-terminal somatomedin-like (SMB) domain of Prg4 is necessary for binding: SMB-deleted Prg4 (DSMB) failed to co-precipitate with GPR146 whereas HEM-deleted Prg4 (DHEM) retained binding. Functional assays using conditioned media demonstrated that WT Prg4 and Δ HEM Prg4 suppressed adipocyte HSL Ser563 phosphorylation, but Δ SMB Prg4 did not (Fig. 5D [↗](#)). These results indicate that Prg4 reduces adipocyte PKA signaling and lipolysis through direct engagement of GPR146 via its SMB domain.

Discussion

This study identifies a liver palmitoylation → hepatokine axis that regulates adipose lipolysis and predisposes to diet-induced obesity. DHHC7 activity in hepatocytes maintains Gai palmitoylation, preserving inhibitory control on adenylyl cyclase. Loss of DHHC7 diminishes Gai palmitoylation, elevates hepatic cAMP–PKA–CREB signaling, and promotes transcriptional upregulation and secretion of Prg4. Circulating Prg4 engages adipocyte GPR146 via its SMB domain, suppresses adipocyte PKA substrate phosphorylation and HSL Ser563 phosphorylation, reduces lipolytic flux, and thereby promotes fat accumulation under HFD. In all, DHHC7 knockout in liver enhances Prg4 expression, and Prg4 interacts with GPR146 in adipose tissue to inhibit lipolysis as described in Fig.5E.

Several aspects of the phenotype are notable. First, D7LKO-driven obesity occurs without overt insulin resistance or hypertriglyceridemia, distinguishing this form of adipose expansion from obesity caused by classical insulin resistance-driven lipotoxicity. Suppression of adipocyte lipolysis by Prg4 likely reduces systemic free fatty acid flux, explaining preserved glucose tolerance despite increased adiposity. Second, elevated plasma cholesterol in D7LKO and Ad-Prg4 mice suggests additional roles for Prg4– GPR146 signaling in hepatic cholesterol handling or intestinal lipid absorption; GPR146 has previously been implicated in cholesterol regulation, consistent with our findings. Third, the identification of Prg4 as a hepatokine expands its known biology beyond joint lubrication and suggests potential broader physiological and pathological roles.

Mechanistically, our data implicate Gai palmitoylation as a critical control point linking hepatic palmitoylation state to transcriptional programs. While the palmitoylation capture assay demonstrates reduced Gai palmitoylation in D7LKO liver, further biochemical work will be needed to define whether Gai is a direct substrate of DHHC7 or whether other DHHC7 targets modulate Gai palmitoylation or membrane targeting. Likewise, the downstream signaling dynamics between hepatic PKA–CREB activation and selective induction of Prg4 versus other CREB targets merit further dissection.

Initially identified as a joint lubricant, Prg4 has been implicated in tissue homeostasis regulation, inflammation as well as [\(41\)](#). In this report, we identified Prg4 as a hepatokine, providing a new insight into Prg4 function.

GPR146 has been implicated in the liver cholesterol metabolism and adipose lipolysis and proposed as a potential therapeutic target for modulating cholesterol homeostasis [\(42\)](#). Our studies here suggest that the Prg4–GPR146 axis provides a testable target: antagonizing Prg4 binding to GPR146, blocking GPR146, or restoring hepatic DHHC7 activity could potentially increase adipose lipolysis and limit diet-induced fat accumulation. Conversely, harnessing Prg4's anti-lipolytic property might be useful where lowering circulating free fatty acids is desirable. Detailed evaluation of organ-specific roles and long-term metabolic consequences will be essential.

In summary, our work links hepatic palmitoylation to hepatokine output and adipose lipolysis, identifying DHHC7 and Prg4–GPR146 as key regulators of diet-induced adiposity. At present, the reason for hypercholesterolemia in D7LKO obese mice is not understood. Two possible scenarios: 1) DHHC7 modulates SREBP-2 in the liver and 2) Prg4 affects GPR146 in the liver, both of which are under investigation.

Materials and Methods

Animals

DHHC7 floxed mice (DHHC7^{F/F} mice) in which the exons 5 and 6 of the DHHC7 gene flanked by loxp sites using the CRISPR-Cas9 technique, were established by Shanghai Model Organisms Center, Inc. The genotyping primers are as following: 5'LoxpF-GTGTGCATTTGAAGTGTCTGCTG, 5'LoxpR - TGCCAACCCGAGAGCTTCCAAG; 3'LoxpF-GATCAGAATGTGGCAGCTTCATC, 3'LoxpR – CACAAGTTAGCCATGGACTGAGG. To generate liver specific inducible DHHC7 knockout mice, DHHC7 floxed mice were crossed with Albumin-CreERT2 (from Shanghai Model Organisms Center, Inc) to generate DHHC7 floxed Albumin-CreERT2 (DHHC7^{F/F}-AlbCREERT2). To delete DHHC7 in liver, DHHC7^{F/F}-AlbCREERT2 mice was oral gavage Tamoxifen (2g/kg, three times, one time for every other day) at six to eight weeks old. The mice were bred and housed at the Hebei University animal facility. Both male and female mice were used for experiments. All animal experiments were approved by the Institutional Animal Care and Use Committees (IACUC) of Hebei University and performed in accordance with all relevant guidelines and regulations of the IACUC of Hebei University. The mice were fed with normal chow diet (NCD) or high fat diet (HFD, 60% fat, Research Diets, Inc., D12492). The body weight were monitored weekly. The food intake was measured every day for individually caged mice for one week. Body composition was determined by nuclear magnetic resonance (Echo MRI). The O₂ consumption, CO₂ production and physical activity were measured in metabolic cages (Columbus Instruments). To examine the adipose tissue growth, different adipose tissues were removed from mice and weighted. Then, the adipose tissues were dissected and fixed in 10% neutral buffered formalin and sectioned for H&E Staining, which was performed in the facility core of Zhejiang Chinese Medical University. For glucose tolerance test (GTT), glucose was intraperitoneally injected at 1g/kg body weight after 12 hours fasting. For insulin tolerance test (ITT), human insulin were intraperitoneally injected at 0.75U/kg body weight after 6 hours fasting. Glucose level was determined with tail blood at indicated time after glucose or insulin injection using a One Touch ultra2 glucose meter as described (12). Serum triglycerides, total cholesterol, high density lipoprotein cholesterol, low density lipoprotein cholesterol and Prg4 were measured with triglyceride determination kit (A110-1-1, Nanjing Jiancheng Bioengineering Company), total cholesterol assay kit(A111-1-1, Nanjing Jiancheng Bioengineering Company), high-density lipoprotein cholesterol assay kit(A112-1-1, Nanjing Jiancheng Bioengineering Company), low-density lipoprotein cholesterol assay kit(A113-1-1, Nanjing Jiancheng Bioengineering Company),Prg4 ELISA kit (RK09131, ABclonal).

Reagents and antibodies

All common chemicals, buffers were from Sigma, Thermo fisher scientific, Cayman biochemical and MCE. The antibodies used in the present study are as the following Anti-Flag (Cat#F3165) and anti-Flag M2-Affinity gel (Cat#A4596) from Sigma. Anti-HSP90 (Cat#4877), Anti-Phospho-HSLSer563 (Cat#4139), Anti-HSL (Cat#4107), Anti-Phospho-PKA Substrate (Cat#9624), Anti-Phospho-CREB Ser133 (Cat#9198), Anti-CREB(Cat#9197), from Cell Signaling Technology. Anti-FASN (Cat#ST1549) from Millipore. Anti-DHHC7 (Cat# A17981), Anti-Prg4(Cat#A15379), from Abcolonal.

Plasmid and viral vector construction and production

The Flag-GPR146 was purchased from SinoBiological(Cat HG12959-CF). To generate lentiviral vector expressing Prg4, lentiviral vector pLV-GFP was digested with SalI, and blunted by T4 DNA polymerase, then digested by BamHI. Prg4 cDNA was amplified by PCR with primers: 5'-

GAAGATCTATGGGGTGGAAAATACTTCC-3' and 5'-TCAAGGACAGTTGTACCAGA TTTTG-3', and the PCR product was digested with BglII. Then, the digested Prg4 cDNA was ligated with the above digested pLV-GFP.

The generation of mutant Prg4 in which either SMB or HEM was deleted were carried out with overlapping PCR with following corresponding primers: SMB domain deletion: Forward, 5'-GAAGA TCTATGGGGTGGAAAATACTTCC and backward, 5'-TGGTGATGTG GGATTTTGAGATGAAACCTGTTG-3', Forward:5'-AATCCCACATCACCATCTC-3' and backward,5'-TCAAGGACAGTTGTACC AGATTTTG-3'; HEM domain deletion: Forward, 5'-GAAGATCTATGGGGTGGAAAATACTTCC-3' and backward, 5'-GAG ATGGTGATGTGGGATT-3'. Forward, 5'-AATCCCACATCACCATCTC-3' and backward, 5'-TCAAGGACAGT TGTACCAGATTTTG-3'. To generate adenoviral vector encoding Prg4, its PCR product was digested by BglII, and inserted into BglII and EcoRV of pAdtrackTBG. The adenoviruses were produced and purified by a CsCl gradient as described (43).

Cell cultures and Transient transfection

HEK293T were grown in high glucose Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% (v/v) fetal bovine serum(FBS), 2mM L-glutamine, 100 units/ml penicillin and 100 µg/ml streptomycin (Invitrogen). The cells were transiently transfected with lipofectamine 2000(Invitrogen) according to manufacturer's instruction.

To produce the culture medium containing different form of Prg4, empty lentiviral vector or lentiviral vectors encoding different forms of Prg4 were transiently transfected into 293T cells. 36 hours posttransfection, the cell culture medium was collected and filtered through 0.45µm syringe filter.

Adipocyte Differentiation

For the 3T3-L1 preadipocyte differentiation (44), 3T3-L1 preadipocytes were cultured for two additional days after reaching 100% confluence and treated with differentiation medium (Dulbecco's modified Eagle's medium–high glucose containing 10% fetal bovine serum, 2.5 µg/ml insulin, 0.5 mM IBMX, 2.5 mM dexamethasone, 2 mM L-glutamine, 100 units/ml penicillin, and 100 µg/ml streptomycin) for 4 days. Then, the medium was changed to regular medium. After 7 days of differentiation, the adipocytes were used for experiments.

For the brown preadipocytes differentiation: the interscapular brown fat pad was dissected from the new born mice(day 1) and digested for 40 minutes with collagenase solution at 37°C. Then, digested tissue were filtered through 100 µm filters and spin at 1000rpm for 10 minutes. The pellet is resuspended and cultured in DMEM-high glucose, 20% FBS. When the cells reached 90% confluence, they were treated with differentiation medium (DMEM-high glucose with 10% FBS, 1µg/mL insulin, 1nmol/L T3, 0.125 mmol/L indomethacin, 1µmol/L dexamethasone, 0.5mmol/L isobutylmethylxanthine and 1µmol/L rosiglitazone) for three days, then switched to DMEM plus 10% FBS, 1µg/mL insulin, and 1nmol/L T3 for four days. After 7 days of differentiation, the adipocytes were used for experiments.

For beige adipocyte differentiation, primary preadipocytes were isolated from 6 to 8 weeks old C57BL6/J mice through collagenase digestion. The differentiation was performed by culturing confluent cells in medium (DMEM-high glucose with 10% FBS, 5µg/mL insulin, 1nmol/L T3, 0.125 mmol/L indomethacin, 1µmol/L dexamethasone, 0.5mmol/L isobutylmethylxanthine and 1µmol/L rosiglitazone) for three days, then switched to DMEM plus 10% FBS, 5µg/mL insulin, and 1nmol/L T3 for four days. After 7 days of differentiation, the adipocytes were used for experiments.

To examine the impact of Prg4 on HSL phosphorylation at Ser563, the differentiated adipocytes were incubated with DMEM or DMEM containing Prg4 for 4 hours.

Western blot

After treatments, cells were washed twice with PBS and extracted with cell lysis buffer (20 mM Tris pH 7.6, 150 mM NaCl, 0.5 mM EDTA, 0.5 mM DTT, 10% glycerol, protease and phosphatase inhibitors). For tissue extracts, the proper amount tissues (i.e. liver etc.) were cut into small pieces

and resuspended in SDS-UREA buffer (50mM Tris, 1%SDS, 10% glycerol, 5mM $\text{Na}_4\text{P}_2\text{O}_7$, 50mM NaF, 5M urea, 5mM EDTA composition). Then, the tissues were homogenized and centrifuged at 13krpm for 10 mins. For total cell lysate and tissue extracts, the protein concentration was determined by BCA kit (Pierce 23225). For cellular fractionation experiments, the cellular fractions were directly dissolved into lysis buffer. Equal amounts of protein were subjected to SDS PAGE electrophoresis and transferred to nitrocellulose membranes (Biorad). After blocking in 5% dry milk, the membranes were incubated with each primary antibody, followed by incubation with a horseradish peroxidase-conjugated secondary antibody. The protein bands were visualized using the ECL detection system (Pierce). The quantification of Western blot was determined with Image J software.

Protein Palmitoylation Assay

The protein palmitoylation assay has been described (45). Briefly, total cytoplasmic extracts were prepared from cultured cells or mouse tissues. Then the extracts were incubated with blocking buffer (100 mM HEPES, pH 7.6, 2.5% SDS, 1 mM EDTA, 1 μ L/ml methyl methanethiosulfonate) for 20 min at 42°C, and the proteins were precipitated with 3 volumes of 100% acetone. After centrifugation the protein pellets were dissolved into binding buffer (100mM HEPES, pH 7.6, 1 mM EDTA) plus thiopropyl-Sepharose beads and incubated for 4 h. The beads were washed three times with binding buffer. Then the proteins on the Sepharose beads were eluted with 100 mM DTT and analyzed by western blot with appropriate antibodies.

Real-time PCR

Total RNA was isolated from mouse tissues or cultured cell by using RNeasy kit (Qiagen), and quantified with a Nanodrop spectrophotometer (ThermoScientific, Wilmington, DE). First Strand cDNA was synthesized with the SuperScript® III First-Strand Synthesis System with random decamers as primers (ThermoFisher). Gene expression was determined by real-time quantitative PCR using a Prism HT7900 instrument with SYBR Green chemistry (Applied Biosystems, Foster City, CA). The primers were indicated in the following: DHHC7F-CTCTGATTCTCTGTGGGCTTC, DHHC7R-GCCTCTCGATCTCTGTTTCATC; DHHC3F-CCACTTCCTGCATTGCTTTG, DHHC3R-TAGCCCATCTTCTCTTCTT CCT; 36BP4F-CTGAGTACACCTTCCCACTTAC, 36BP4R-CCGAATCCCATATCCTCATCTG. The procedure was as 94°C for 30s, 60°C for 30s, and 72°C for 30s for 40 cycles.

Proteomics analysis

The liver tissue and plasma were harvested from the high fat diet male mice, then were performed to quantitative proteomics analysis at Shanghai OE Biotech Co., Ltd. (China). The thresholds of fold change (>1.2 for plasma proteomics, >1.5 for liver tissue proteomics) and P-value (<0.05) were used to identify differentially expressed proteins (DEPs).

Immunoprecipitation assays

Transfected HEK293 cells were extracted with immunoprecipitation buffer (150 mM NaCl, 25 mM Tris, pH 7.6, 0.5 mM EDTA, 10% glycerol, 0.5% NP-40 plus a protease inhibitor mixture). 500 ug of total proteins from the cell lysates were subjected to immunoprecipitation with the corresponding antibodies.

Statistical Analyses

The quantification of Western blot is carried out with Image J. Means $\pm$ S.D. were calculated, and statistically significant differences among groups were determined by one-way analysis of variance followed by post hoc comparisons or by two-tailed unpaired Student's *t*-tests as appropriate, with significance set at $p < 0.05$.

Data availability

Figure 1-source data 1 and Figure 2-source data 1 contain the numerical data used to generate the figures.

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Additional files

[Supplementary file 1](#) 

Additional information

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this study, the authors' aim was to determine whether hepatic palmitoylation is a physiologically relevant regulator of systemic metabolism. The data demonstrate that loss of DHHC7 in hepatocytes disrupts Gαi palmitoylation, enhances cAMP-PKA-CREB signaling, and drives transcriptional upregulation and secretion of Prg4. The KO mice display increased body weight, fat mass, and plasma cholesterol, but at 12 weeks on HFD, do not exhibit insulin resistance. The potential mechanism underlying the metabolic phenotype was examined by assessing adipocyte signaling and by exploring whether Prg4 acts through GPR146. Through this pathway, the authors intend to link DHHC7-dependent palmitoylation to the regulation of hepatokines that exert systemic metabolic effects.

Strengths:

(1) Hepatic palmitoylation in systemic metabolic regulation is largely unexplored. The authors demonstrate the role of DHHC7 in vivo using a successful liver-specific knockout mouse model that causes HFD-dependent obesity without insulin resistance.

(2) Several studies were performed on chow and HFD, as well as male and female mice.

- (3) Plasma proteomics identified Prg4 as a circulating factor elevated in KO mice. Prg4 overexpression phenocopied the KO mice.
- (4) There is solid mechanistic data supporting the hypothesis that hepatic DHHC7 loss selectively increases Prg4 secretion as a hepatokine.
- (5) There is convincing evidence for the DHHC7 mechanism in liver: DHHC7 controls cAMP-PKA-CREB via Gai palmitoylation. The authors recognize that the palmitoylation change is causative rather than correlated, and this needs to be more fully explored in the future.
- (6) Strong in vitro data support that Prg4 acts through adipocyte GPR146 via its SMB domain

Weaknesses:

- (1) The assessment of liver and adipose tissue responses to DHH7 loss is insufficient to support claims that it alters systemic lipolysis. In this new mouse model, liver histology is necessary, especially given the cholesterol increase in the KO. As this is a newly established mouse line, common assessments of the liver during HFD feeding would be important for interpreting the phenotype.
- (2) The data show DHH7 loss causes adipose tissue dysfunction and alterations in lipid metabolism. Beyond that, I suggest not stating more regarding the phenotype of the DHH7 mice for this work. A thorough analysis would be needed to determine which factor drives the obesity and changes in energy balance in the mice. For example, the KO mice had lower oxygen consumption (but no change in CO₂ production, which is also usually similarly altered), suggesting a CNS component could drive obesity. However, since the data are not normalized for lean mass and there is no information about locomotor activity, this analysis is incomplete. RER may be informative if available. A broad conservative description of the KO phenotype would be more accurate since Pgr4 has many paracrine targets and likely has autocrine signaling in the liver.
- (3) Most references to lipolysis or lipolysis flux systemically would be inaccurate. To suggest a suppression of lipolysis, serum NEFA would need to be measured, and in vivo or in vitro lipolysis assays performed to test the effect of DHH7 loss or the specificity of PGR4 action on adipocytes in vivo. To demonstrate adipose tissue dysfunction, analysis of lipogenesis markers, canonical markers for insulin sensitivity, and mitochondrial dysfunction should be performed/measured.
- (4) Line 179: The experiment was performed in brown adipocytes to show that Prg4 does not affect p-CREB Figure S8 under the heading: "DHHC7 controls hepatic PKA-CREB activity through Gai palmitoylation to regulate Prg4 transcription." Unless repeated using liver lysate, the conclusions stated in the text throughout the paper should be revised.
- (5) It appears that the serum and liver proteomics were only assessed for factors that increased in KO mice? Were proteins that were significantly decreased analyzed?
- (6) The beige adipocyte culture method is unclear. The methods do not describe the fat pad used, and the protocol suggests the cells would be differentiated into mature white adipocytes. If they are beige cells, a reference for the method, gene expression, and cell images could support that claim.
- (7) The use of tamoxifen can confound adipocyte studies, as it increases beigeing and weight gain even after a brief initiation period. Both groups were treated with Tam, but another way to induce Cre would be ideal.
- (8) Evidence for the lack of the glucose phenotype is incomplete. One reason could be due to the IP route of glucose administration, which has a large impact on glucose handling during a

GTT. To confirm the absence of a glucose tolerance phenotype, an OGTT should be performed, as it is more physiological. In addition, the mice should be fed for 16 weeks. Prg4 affects immune cells, changing how adipose tissue expands, and 12 weeks of HFD feeding is often not long enough to see the effects of adipose tissue inflammation spilling over into the system.

(9) There may be liver-adipose tissue crosstalk in KO mice, but this was not fully assessed in this study and would be difficult to determine in any setting, given the diverse cell types that are targets of Pdg4. The crosstalk claim is unnecessary to share the basic premises; there is the DHH7 mechanism/phenotype and the Pgr4 mechanism/phenotype, and while there is no Pgr4 adipose direct mechanism, the paper can be successfully reframed.

(10) Although the DHH7 loss on the chow diet did not result in a phenotype, did the Pgr4 increase in the KO mice on chow? This would determine whether either i) the expression of Pgr4 is dependent on HFD/obesity, or ii) circulating Pgr4 has effects only in an HFD condition. The receptors may also change on HFD, especially in adipocytes.

Impact:

This work would significantly contribute to the study of liver metabolism, provided it includes data describing the liver. The role of Pgr4 in adipocytes and other cell types is of substantial value to the field of metabolism. By reframing the paper and conducting some key experiments, its quality and impact can be increased.

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Reviewer #2 (Public review):

In the current report, Sun and Colleagues sought to determine the liver-specific role that DHHC7, a DHHC palmitoyltransferase protein, plays in regulating whole-body energy balance and hepatic crosstalk with adipose tissues. The authors generated an inducible, liver-specific DHHC7 knockout mouse to determine how altered palmitoylation in hepatocytes alters hepatokine production/secretion, and in turn, systemic metabolism. The ablation of DHHC7 was found to alter the production of proteoglycan 4 (Prg4), a hepatokine previously linked to metabolic regulation. The authors propose that the change in Prg4 production is mediated by the loss of Gai palmitoylation, due to DHHC7 ablation, thereby augmenting cAMP-PKA-CREB signaling in hepatocytes, which alleviates the 'brake' on Prg4 production. The authors further propose that Prg4 overexpression leads to excessive binding to GPR146 on adipocytes, which in turn suppresses PKA-mediated HSL activation, promoting impairments in lipolysis, leading to obesity. The report is interesting and generally well-written, but it appears to have some clear gaps in additional data that would aid in interpretation. The addition of confirmatory culture studies would be incredibly helpful for testing the hypotheses being explored. My comments, concerns, and/or suggestions are outlined below in no particular order.

(1) Figures: All data should be presented in dot-boxplot format so the reader knows how many samples were analyzed for each assay and group. n=3 for some assays/experiments is incredibly low, particularly when considering the heterogeneity in responsiveness to HFD, food intake, etc....

(2) Figure 1E-F: It is unclear when the food intake measure was performed. Mice can alter their feeding behavior based on a myriad of environmental and biological cues. It would also be interesting to show food intake data normalized to body mass over time. Mice can counterregulate anorexigenic cues by altering neuropeptide production over time. It is not clear if this is occurring in these mice, but the timing of measuring food intake is important. Additionally, the VO2 measure appears to be presented as being normalized to total body mass, when in fact, it would probably be more accurate to normalize this to lean body mass.

Normalizing to total body mass provides a denominator effect due to excessive adiposity, but white fat is not as metabolically active as other high-glucose-consuming tissues. If my memory serves me right, several reports have discussed appropriate normalizations in circumstances such as this.

(3) Figure 1J-N: It is not all that surprising that fasting glucose and/or TGs were found to be similar between groups. It is well-established that mice have an incredible ability to become hyperinsulinemic in an effort to maintain euglycemia and lipid metabolism dynamics. A few relatively easy assays can be performed to glean better insights into the metabolic status of the authors' model. First, fasting insulin concentrations will be incredibly helpful. Secondly, if the authors want to tease out which adipose depot is most adversely affected by ablation, they could take an additional set of CON and KO mice, fast them for 5-6 hours, provide a bolus injection of insulin (similar to that provided during an insulin tolerance test), and then quickly harvest the animals ~15 minutes after insulin injections; followed by evaluating AKT phosphorylation. This will really tell them if these issues have impairments in insulin signaling. The gold-standard approach would be to perform a hyperinsulinemic-euglycemic clamp in the CON and KO mice. I now see GTT and ITT data, but the aforementioned assays could help provide insight.

(4) Figure 3A: This looks overexposed to me.

(5) Figures 3-4: It appears that several of these assays could be complemented with culture-based models, which would almost certainly be cleaner. The conditioned media could then be used from hepatocyte cultures to treat differentiated adipocytes.

(6) Figure 4: It is unclear how to interpret the phospho-HSL data because the fasting state can affect this readout. It needs to be made clear how the harvest was done. Moreover, insulin and glucagon were never measured, and these hormones have a significant influence over HSL activity. I suspect the KO mice have established hyperinsulinemia, which would likely affect HSL activity. This provides an example of why performing some of these experiments in a dish would make for cleaner outcomes that are easier to interpret.

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Reviewer #3 (Public review):

Summary:

In the current manuscript, Sun et al aimed to determine the metabolic function of hepatocyte DHHC7, one of the key enzymes in protein palmitoylation. They generated inducible liver-specific *Dhhc7* knockout mice and discovered that *Dhhc7*-LKO mice are more prone to gain weight and develop adipose expansion and obesity. Via unbiased proteomic analysis, they identified PRG4 as one of the top secreted factors in the liver of *Dhhc7*-LKO mice. Hepatic overexpression of PRG4 recapitulates the obesity phenotype observed in *Dhhc7*-LKO mice. At the mechanistic level, PRG4, once secreted from the liver, can bind to GPR146 on adipocytes and inhibit PKA-HSL signaling and lipolysis. Taken together, their findings suggest a novel pathway by which the liver communicates with adipose tissue and impacts systemic metabolism.

Strengths:

(1) The systemic metabolic homeostasis depends on coordination among metabolically active tissues. Thus, active communication between the liver and adipose tissue when facing nutritional challenges (such as high-fat diet feeding) is crucial for achieving metabolic health. The concept that the liver can communicate with adipose tissue and impact the lipolysis process via secreted hepatokines is quite significant but remains poorly understood.

- (2) Hepatocyte Dhhc7 knockout mice developed a significant obesity phenotype, which is associated with adipose expansion.
- (3) Unbiased proteomic analysis identified PRG4 as one of the top secreted factors in the liver of Dhh7-LKO mice. Hepatic overexpression of PRG4 recapitulates the obesity phenotype observed in Dhh7-LKO mice.
- (4) In vitro cell-based assay showed that PRG4 can bind to adipocyte GPR146, inhibit PKA-mediated HSL phosphorylation, and subsequently, the lipolysis process.

Weaknesses:

- (1) Lack of a causal-effect study to generate evidence directly linking hepatocyte DHH7 and PRG4 in driving adipose expansion and obesity upon HFD feeding.
- (2) Lack of direct evidence to support that PRG4 inhibits adipocyte lipolysis via GPR146. A functional assay demonstrating adipocyte lipolysis is required.
- (3) The conclusion is largely based on the correlation evidence.

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Author response:

Public reviews:

Reviewer #1 (Public review):

Weaknesses:

(1) The assessment of liver and adipose tissue responses to DHH7 loss is insufficient to support claims that it alters systemic lipolysis. In this new mouse model, liver histology is necessary, especially given the cholesterol increase in the KO. As this is a newly established mouse line, common assessments of the liver during HFD feeding would be important for interpreting the phenotype.

We will add the data of the liver histology in the revised version.

(2) The data show DHH7 loss causes adipose tissue dysfunction and alterations in lipid metabolism. Beyond that, I suggest not stating more regarding the phenotype of the DHH7 mice for this work. A thorough analysis would be needed to determine which factor drives the obesity and changes in energy balance in the mice. For example, the KO mice had lower oxygen consumption (but no change in CO₂ production, which is also usually similarly altered), suggesting a CNS component could drive obesity. However, since the data are not normalized for lean mass and there is no information about locomotor activity, this analysis is incomplete. RER may be informative if available. A broad conservative description of the KO phenotype would be more accurate since Pgr4 has many paracrine targets and likely has autocrine signaling in the liver.

We will add the data of CO₂ production, locomotor activity and RER in the revised version.

(3) Most references to lipolysis or lipolysis flux systemically would be inaccurate. To suggest a suppression of lipolysis, serum NEFA would need to be measured, and in vivo or in vitro lipolysis assays performed to test the effect of DHH7 loss or the specificity of PGR4 action on adipocytes in vivo. To demonstrate adipose tissue dysfunction, analysis of lipogenesis markers, canonical markers for insulin sensitivity, and mitochondrial dysfunction should be performed/measured.

We will measure the serum NEFA to test the effect of DHH7. We will analyze the lipogenesis markers, canonical markers for insulin sensitivity, and mitochondrial dysfunction.

(4) Line 179: The experiment was performed in brown adipocytes to show that Prg4 does not affect p-CREB Figure S8 under the heading: "DHH7 controls hepatic PKA-CREB activity through Gai palmitoylation to regulate Prg4 transcription." Unless repeated using liver lysate, the conclusions stated in the text throughout the paper should be revised.

The figure S8 is to demonstrate that Prg4 has no impact on forskolin induced CREB phosphorylation at Ser133, and provide the evidence that the prg4 acts on the upstream of adenylyl cyclase. We will revise the description.

(5) It appears that the serum and liver proteomics were only assessed for factors that increased in KO mice? Were proteins that were significantly decreased analyzed?

We are analyzing the decreased proteins in the following project.

(6) The beige adipocyte culture method is unclear. The methods do not describe the fat pad used, and the protocol suggests the cells would be differentiated into mature white adipocytes. If they are beige cells, a reference for the method, gene expression, and cell images could support that claim.

We will add a reference for the method, gene expression, and cell images.

(7) The use of tamoxifen can confound adipocyte studies, as it increases beigeing and weight gain even after a brief initiation period. Both groups were treated with Tam, but another way to induce Cre would be ideal.

We will use the Doxycycline-inducible systems in the future.

(8) Evidence for the lack of the glucose phenotype is incomplete. One reason could be due to the IP route of glucose administration, which has a large impact on glucose handling during a GTT. To confirm the absence of a glucose tolerance phenotype, an OGTT should be performed, as it is more physiological. In addition, the mice should be fed for 16 weeks. Prg4 affects immune cells, changing how adipose tissue expands, and 12 weeks of HFD feeding is often not long enough to see the effects of adipose tissue inflammation spilling over into the system.

We will perform the OGTT and feed the mice for 16 weeks in the future.

(9) There may be liver-adipose tissue crosstalk in KO mice, but this was not fully assessed in this study and would be difficult to determine in any setting, given the diverse cell types that are targets of Pdg4. The crosstalk claim is unnecessary to share the basic premises; there is the DHH7 mechanism/phenotype and the Pgr4 mechanism/phenotype, and while there is no Pgr4 adipose direct mechanism, the paper can be successfully reframed.

We will reframe the paper.

(10) Although the DHH7 loss on the chow diet did not result in a phenotype, did the Pgr4 increase in the KO mice on chow? This would determine whether either i) the expression of Pgr4 is dependent on HFD/obesity, or ii) circulating Pgr4 has effects only in an HFD condition. The receptors may also change on HFD, especially in adipocytes.

We will test the Prg4 in the KO mice on chow diet.

Reviewer #2 (Public review):

(1) Figures: All data should be presented in dot-boxplot format so the reader knows how many samples were analyzed for each assay and group. $n=3$ for some assays/experiments is incredibly low, particularly when considering the heterogeneity in responsiveness to HFD, food intake, etc.

We will present the data in dot-boxplot format.

(2) Figure 1E-F: It is unclear when the food intake measure was performed. Mice can alter their feeding behavior based on a myriad of environmental and biological cues. It would also be interesting to show food intake data normalized to body mass over time. Mice can counterregulate anorexigenic cues by altering neuropeptide production over time. It is not clear if this is occurring in these mice, but the timing of measuring food intake is important. Additionally, the VO₂ measure appears to be presented as being normalized to total body mass, when in fact, it would probably be more accurate to normalize this to lean body mass. Normalizing to total body mass provides a denominator effect due to excessive adiposity, but white fat is not as metabolically active as other high-glucose-consuming tissues. If my memory serves me right, several reports have discussed appropriate normalizations in circumstances such as this.

We will see how to be more accurate to normalize.

(3) Figure 1J-N: It is not all that surprising that fasting glucose and/or TGs were found to be similar between groups. It is well-established that mice have an incredible ability to become hyperinsulinemic in an effort to maintain euglycemia and lipid metabolism dynamics. A few relatively easy assays can be performed to glean better insights into the metabolic status of the authors' model. First, fasting insulin concentrations will be incredibly helpful. Secondly, if the authors want to tease out which adipose depot is most adversely affected by ablation, they could take an additional set of CON and KO mice, fast them for 5-6 hours, provide a bolus injection of insulin (similar to that provided during an insulin tolerance test), and then quickly harvest the animals ~15 minutes after insulin injections; followed by evaluating AKT phosphorylation. This will really tell them if these issues have impairments in insulin signaling. The gold-standard approach would be to perform a hyperinsulinemic-euglycemic clamp in the CON and KO mice. I now see GTT and ITT data, but the aforementioned assays could help provide insight.

We have the data for evaluating AKT phosphorylation and will add it in the revised version.

(4) Figure 3A: This looks overexposed to me.

We will replace it with short exposed one.

(5) Figures 3-4: It appears that several of these assays could be complemented with culture-based models, which would almost certainly be cleaner. The conditioned media could then be used from hepatocyte cultures to treat differentiated adipocytes.

We will perform the cell culture experiments for Figures 3-4

(6) Figure 4: It is unclear how to interpret the phospho-HSL data because the fasting state can affect this readout. It needs to be made clear how the harvest was done. Moreover, insulin and glucagon were never measured, and these hormones have a significant influence over HSL activity. I suspect the KO mice have established hyperinsulinemia, which would likely affect HSL activity. This provides an example of why performing some of these experiments in a dish would make for cleaner outcomes that are easier to interpret.

We will perform some experiments in cell culture dish.

Reviewer #3 (Public review):*Weaknesses:*

(1) Lack of a causal-effect study to generate evidence directly linking hepatocyte DHH7 and PRG4 in driving adipose expansion and obesity upon HFD feeding.

We will perform the causal-effect study to demonstrate the hypothesis.

(2) Lack of direct evidence to support that PRG4 inhibits adipocyte lipolysis via GPR146. A functional assay demonstrating adipocyte lipolysis is required.

We will add the direct evidence in the revised version.

(3) The conclusion is largely based on the correlation evidence.

We will perform the experiment to strengthen the conclusion base on the a causal-effect study.

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